



Research paper

Acaricidal activity and clinical safety of arecoline hydrobromide on calves infested with cattle tick *Rhipicephalus microplus* (Acari: Ixodidae)

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ABSTRACT

The objective of the present study was to determine the acaricidal activity of arecoline hydrobromide against cattle tick *Rhipicephalus microplus* infesting calves. *in vivo* efficacy of arecoline emulsified with polysorbate-80 (2%) was evaluated using ear bag method with the effective dose of 12.5 mg/mL applied on ear pinna of calves infested with ticks. Control group received polysorbate-80 (2%) after larvae infestation, however, reference group received deltamethrin (0.5 %). The experiment was continued for six days (144 h) and treatment of drug was given twice a day. Daily observation of calves was done to count the number of ticks after treatment. Acute dermal toxicity study for test drug was performed on wistar rats. Clinical safety of arecoline was determined by examining hematological profile and skin irritancy assay for calves infested with ticks. Results showed that arecoline significantly ($p < 0.01$) reduced the number of ticks attached to ear pinna of calves. Fewer number of ticks remained on calves skin at 120 h and 144 h were 8.09 and 6.21, respectively after treatment with arecoline in comparison to control group. Treatment of animals with arecoline hydrobromide significantly ($p < 0.01$) restored the hematological profile of animals as hemoglobin (Hb) level was 9.01 g/100 mL, PVC was 29.24 %, TEC and TLC were 5.23 and 7.19 10^6 /cumm, respectively as compared to the control group having Hb 9.48 g/100 mL, PVC 31.60 %, TEC 5.64 10^6 /cumm and TLC 7.27 10^6 /cumm. Arecoline showed no toxicity while applied on wistar rats. The drug was mild irritative for an initial 20 min to the calves after that no redness or erythema was seen on the skin of the animals. Thus, arecoline hydrobromide may be an effective alternative to be used as herbal ectoparasiticide for the eradication of *R. microplus* ticks.

1. Introduction

Cattle ticks are the well-known ectoparasites that act as vector for many pathogens affecting the health of animals (Jain et al., 2020a,b; Souza et al., 2018). Cattle are the fundamental asset for the livelihood of farmers and poor rural communities in India (Jain et al., 2020c; Biradar et al., 2013). *Rhipicephalus microplus* (Acari: Ixodidae) a cattle tick is considered to be one of the major threats as an external parasite that affecting the cattle industry around the world (Gomes and Neves, 2018). *R. microplus* has a wide dispersal capability that is facilitated by the movement of animals (Sungirai et al., 2018). The studies on the epidemiology of *R. microplus* revealed that they are capable of producing 2–3.5 generations annually and responsible for transmitting diseases like anaplasmosis and babesiosis (Miraballes and Riet-Correa, 2018). The eradication and control of ticks are mostly based on the use of conventional synthetic acaricides and organophosphate compounds.

These substances have shown a toxic effect on both cattle and workers (Vudriko et al., 2018). Additionally, frequent use of these chemicals causes resistance in the tick population (Rodriguez-Vivas et al., 2018). Resistance to ixodicides by ticks has been very prevalent in tropical and subtropical regions that adding the problem of livestock industry (Aguilar et al., 2018). Thus, research is still going on to develop new drugs that are effective and used as an alternative drug to the existing acaricidal drug for which ticks have developed resistance.

Herbal medicines and their phytoconstituents have potential for the development of effective acaricidal drugs (Afrin et al., 2018). The bioactive compounds of medicinal plants are responsible for their therapeutic efficacy (Mustafa et al., 2017). *Areca catechu* L. is a popular traditional medicine chewed by the people for killing worms in the abdominal cavity (Peng et al., 2015). It belongs to the family Arecaceae found abundantly in Southeast Asia. *A. catechu* has the potential to treat diseases such as depression, parasitic disease, jaundice, dyspepsia,

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diarrhea, and disorders of the digestive system. One of the studies reported that chewing of *A. catechu* may reduce epilepsy attacks in affected people (Mateen et al., 2017). Researchers reported that the clinical adverse events of *A. catechu* were less and more controllable. As such, clinicians must standardize their experimental workflows and combine *A. catechu* with other herbal drugs for its safe application (Sun et al., 2017). The use of *A. catechu* for medicinal purposes and in solving health-related problems is recommended (Jaiswal et al., 2011). The biological activities of *A. catechu* are due to the presence of alkaloids as major bioactive compounds i.e. arecoline, guvacoline, arecaidine, norarecoline, nicotine, homoarecoline, and isoguvacine (Peng et al., 2015). Other phytoconstituents found in *A. catechu* are isorhamnetin, quercetin, luteolin, chryseriol, liquiritigenin, jacareubin, catechin, epicatechin, procyanidins, arecatannins, arborinol, ursonic acid, fernenol, cycloartenol, and fatty acids.

Arecoline is an effective phytoconstituent of *A. catechu* and predominantly found in Areca nut that is responsible for various pharmacological effects including anti-parasitic activity and useful in disorders of nervous, cardiovascular, endocrine and digestive system (Liu et al., 2016). Arecoline prevents the growth of pre-adipocytes and pathways of reactive oxygen species (Tian et al., 2018). *in vitro* studies have shown that arecoline possess acaricidal activity against *R. microplus* (Bigg and Purvis, 1976). Arecoline as acetylcholine agonist can reduce oviposition and completely abolish it in high doses (Booth, 1989). Arecoline is rapidly eliminated from rat plasma within 45 min after its oral administration at a dose of 150 mg/kg (Pan et al., 2017). However, the short half-life of arecoline in the blood and transdermal or topical dosage forms may avoid the absorption variability (Wu et al., 2014). Herbal oils and plant compounds are effective in controlling *R. microplus* ticks on animals (Valente et al., 2017; Chagas et al., 2014). Furthermore, topical delivery of drug offer several advantages like maintain sustained release, bypass first-pass metabolism, reduced systemic effects, decrease in dosing frequency and termination of medication (Paudel et al., 2010). Therefore, in the present study arecoline hydrobromide was applied topically to the calves to evaluate its acaricidal efficacy. The acute dermal toxicity study was performed on wistar rats before application on calves. The safety concerns of arecoline were achieved by analyzing the hematological profile of calves blood and skin irritation assay.

2. Material and methods

2.1. Drugs and chemicals

Arecoline hydrobromide and amitraz were procured from Cayman Chemical Pvt. Ltd., New Delhi. Deltamethrin and polysorbate-80 were purchased from Sigma Aldrich, USA. Other chemicals and reagents used in the study were of analytical grade, all are commercially available with purified to 99 % purity.

2.2. Collection and identification of ticks

Ticks were collected by direct isolation from cattle infested naturally found in the local cattle sheds located at Bhilai (21.1569 °N, 81.2656 °E), Chhattisgarh, India in a wide mouth container covered with a muslin cloth. Live adult ticks both male and female were picked with forceps and identified under a stereomicroscope based on the morphological characteristics of *R. microplus* according to the method of Soulsby, 1982. The collected ticks (engorged females) were kept under laboratory conditions (Temperature 27 ± 2 °C and 80 ± 5 % relative humidity). Ticks were fed with pre-warmed (37 °C) blood given through goldbeater's skin as a feeding membrane (Trentelman et al., 2017).

2.3. *In vivo* evaluation of the acaricidal activity of arecoline hydrobromide

2.3.1. Acute dermal toxicity study

Acute toxicity study was performed according to OECD guideline 402. Five female Wistar rats were selected for the study to evaluate dermal toxicity of AH (12.5 mg/mL). The animals were procured from central animal house of the institute. The experimental protocol was approved by Institutional Animal Ethical Committee (IAEC), Columbia Institute of Pharmacy, Raipur. The dorsal hair of animals with total area of 5 cm² was removed to apply the extract. Emulsifiable concentrate of AH made with polysorbate-80 (2%) was applied on the shaved area of the animal's skin. The contact time of formulation with skin was enhanced by placing porous gauze dressing and non-irritating tape on the wounded area for 24 h. The test compounds were removed after exposure of 24 h with the help of warm water or sunflower oil. The animals were observed daily over the cage but weight of the animals was noted weekly for 14 days. The skin was observed for swelling, inflammation, redness, irritation and other physiological and anatomical changes after 4 h of the application of ointment followed by 14 days observation.

2.3.2. Ear bag method

2.3.2.1. Experimental design. Indigenous calves (red Sahiwal breed) free from ticks, older than six months were selected for the study. Total eighteen animals were selected for the study and divided into three groups i.e. Control group received polysorbate-80 after larvae infestation, reference group received emulsifiable concentrate of deltamethrin (0.5 %) topically and test group was treated topically with 12.5 mg/mL emulsifiable concentrate of arecoline hydrobromide. The animals were kept separately in an individual tick-proof shed with dimension 2.7 m × 2.5 m. The shed was ventilated and walls and floors were free from crevices. Regular monitoring of sheds and calves were performed to maintain healthy and disease-free conditions. The animals were fed with cattle feed mash (Praktikum Agritech LLP, Balod, India) along with paddy straw and providing water *ad libitum*.

2.3.2.2. Procedure. Acaricidal activity of arecoline hydrobromide (12.5 mg/mL) was evaluated by applying it topically on calves skin in the form of emulsifiable concentrate using polysorbate-80 as a surfactant. The selected animals were examined daily to be assured about their healthy condition and providing measured cattle feed, paddy husk and water *ad libitum*. Blood smears were collected periodically to ascertain about the disease free condition of animals. These experimental condition was maintained for animals since their birth to six months before starting the experiment. Animals were fed with 120–140 unfed *R. microplus* larvae (14 days old). Ear pinna of calves were shaved and larvae were placed on it using ear bags (12 cm × 16 cm). The ear bags were tied on base of ear pinna. The seed ticks were counted after 24 h, which was placed on ear pinna of animals and assured that an average of 35 ticks should attach to the pinna of each treated calves. The treatment was continued for 6 days and ticks attached to the pinna were recorded after every 24 h. The criteria of evaluation for this ear bag method was the reduction of number of ticks after treatment with the test drug (Ghosh and Azharianambi, 2007; Zaman et al., 2012).

2.3.3. Hematological studies

Hematological analysis of calves blood was performed according to the method of Schalm et al. (1975). A sterilized needle (18 gauge) was used to collect blood from the jugular vein of calves in vials containing an anticoagulant (EDTA, 10 %). Blood samples were collected initially at 0 h before infestation of ticks and finally at the end of the experiment i.e. 144 h after treatment. The vials were then sealed and labeled properly for further identification. Hematological values were evaluated by

collecting blood before and after treatment as per the standard procedure. The samples were analyzed for Hemoglobin (Hb), Packed cell volume (PCV), Total leucocyte count (TLC), Differential leucocyte count (DLC), Total erythrocyte count (TEC), and platelet count through automatic haemoanalyser (Medonic M20 Cell Counter, Boule Diagnostics AB Domnarvsgatan, Sweden).

2.4. Assay for skin irritation evaluation

The untreated calves ($n = 6$) free from ticks were included in the study for assay of skin irritation. Both the thighs of animals were shaved and marked the test area (16×16 cm). On day 0 the test area was washed with water and allowed to dry at room temperature. The animals were treated with 12.5 mg/mL arecoline hydrobromide emulsifiable concentrate and control group animals were treated with polysorbate-80 (2%) with a rate of 0.5 mL/cm^2 of skin. The erythema was recorded on animals skin at 20 min, 6 h, 12 h, 24 h, and 48 h after treatment and scored according to the Draize system (Draize et al., 1994) as follows: no change assigned as 0; the slight change was assigned as 1; pale redness in the applied area was assigned as 2; definite redness on the applied area was assigned as 3, and crimson red on the applied area was assigned as 4.

2.5. Statistical analysis

Values are represented as mean $\pm$ SD ($n = 6$). Statistical analysis was performed using one-way ANOVA with Bonferroni post-tests through graph pad prism software version 8.0, statistically different at $*p < 0.05$, $**p < 0.01$ and $***p < 0.001$ in comparison to the control group.

3. Result

3.1. Acute dermal toxicity study

AH was found non toxic upon application on the skin of animals. It did not show any type of irritation, swelling, inflammation and any other change on the skin. No any sign of salivation, convulsion, diarrhoea, tremors, lethargy, sleep and coma were seen. Table 1 shows cage side observation of animals treated with test compounds.

Table 1

Toxicity study of arecoline hydrobromide upon topical application on wistar rats.

Response	Time									
	4 h	24 h	1 day	2 days	4 days	6 days	8 days	10 days	12 days	14 days
Alertness	+	+	+	+	+	+	+	+	+	+
Irritability	-	-	-	-	-	-	-	-	-	-
Fearfulness	-	-	-	-	-	-	-	-	-	-
Touch Response	+	+	+	+	+	+	+	+	+	+
Restlessness	-	-	-	-	-	-	-	-	-	-
Abdominal Tone	+	+	+	+	+	+	+	+	+	+
Tremors	-	-	-	-	-	-	-	-	-	-
Writhing	-	-	-	-	-	-	-	-	-	-
Corneal reflexes	+	+	+	+	+	+	+	+	+	+
Diarrhea	-	-	-	-	-	-	-	-	-	-
Food and water intake	+	+	+	+	+	+	+	+	+	+
Respiration rate	+	+	+	+	+	+	+	+	+	+
Pupil reaction to light	+	+	+	+	+	+	+	+	+	+
Skin color & texture	+	+	+	+	+	+	+	+	+	+
Spontaneous activity	+	+	+	+	+	+	+	+	+	+
Heart rate	+	+	+	+	+	+	+	+	+	+
Convulsion	-	-	-	-	-	-	-	-	-	-
Aggressiveness	-	-	-	-	-	-	-	-	-	-
Vomiting	-	-	-	-	-	-	-	-	-	-
Sedation	-	-	-	-	-	-	-	-	-	-
Locomotion	+	+	+	+	+	+	+	+	+	+
Edema	-	-	-	-	-	-	-	-	-	-
Mortality	-	-	-	-	-	-	-	-	-	-

+ Present.

- Absent.

3.2. In vivo efficacy of arecoline hydrobromide

Table 2 shows the effect of arecoline hydrobromide on calves infected with *R. microplus* ticks. Results exhibited that arecoline reduced the number of ticks attached to ear pinna of calves. The number of ticks remained on calves skin at 120 h and 144 h were 8.09 and 6.21, respectively after treatment with arecoline in comparison to the control group. However, the calves were completely tick free on 120 h and 144 h after treatment with the reference drug deltamethrin. Fig. 1 shows the percentage reduction in number of ticks attached to calves skin. The percentage reduction of ticks number after treatment with AH was 82.26 % at 144 h in comparison to control group i.e. 6.21 %. Deltamethrin showed 100 % reduction in number of ticks attached to calves skin on 120 h and 144 h. The control animals were treated with amitraz after the experiment as per the guidelines of WAAVP (Vercruysse et al., 2006).

3.3. Hematological estimation

Table 3 shows the hematological profile of infested calves after treatment with arecoline hydrobromide emulsifiable concentrate at 144 h. Treatment of animals with arecoline hydrobromide restored the hematological profile of animals near to normal values as hemoglobin level

Table 2

Effect of arecoline hydrobromide on number of *R. microplus* ticks attached to calves.

Time (h)	No. of ticks ($n = 35$)		
	Control	Deltamethrin	AH
24	35.00 ± 0.01	$22.85 \pm 2.01^{***}$	28.03 ± 1.31
48	34.98 ± 0.04	$7.45 \pm 1.07^{***}$	26.05 ± 2.01
72	34.95 ± 0.01	$2.66 \pm 1.49^{***}$	$21.22 \pm 2.26^*$
96	33.92 ± 0.05	$1.75 \pm 0.13^{***}$	$13.12 \pm 1.92^{**}$
120	33.02 ± 0.05	0 ***	$8.09 \pm 1.40^{***}$
144	33.56 ± 0.02	0 ***	$6.21 \pm 1.54^{***}$

Data are expressed at mean $\pm$ SD ($n=6$), significantly different at $*p<0.05$, $**p<0.01$ and $***p<0.001$ in comparison to control group. Control group - Polysorbate-80 (2%), Deltamethrin - 0.5% and AH - Arecoline hydrobromide (12.5 mg/ml).

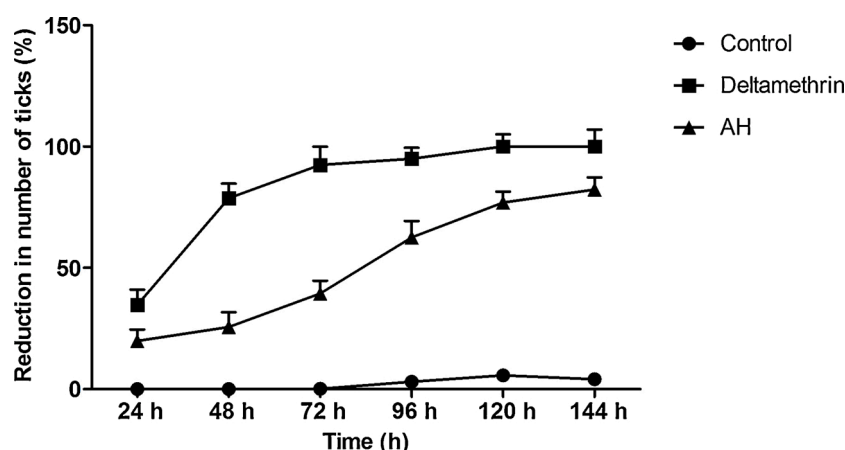


Fig. 1. Effect of arecoline hydrobromide in reducing number of ticks attached to calves skin. Control = water with polysorbate-80 (2%), Deltamethrin (0.5 %), AH = Arecoline hydrobromide (12.5 mg/mL).

Table 3

Hematological values of calves after treatment with arecoline hydrobromide at 144 h.

S. No.	Components	Control	Deltamethrin	AH
1	Hb (g/100 mL)	9.48 ± 0.73	7.44 ± 1.22 ***	9.01 ± 0.11
2	PCV (%)	31.60 ± 0.95	28.22 ± 0.39 *	29.24 ± 0.39
3	TEC (10 ⁶ /cumm)	5.64 ± 0.44	6.21 ± 0.78	5.92 ± 0.31
4	TLC (10 ⁶ /cumm)	7.27 ± 0.32	7.43 ± 0.10	7.19 ± 0.10
	DLC (%)			
	Lymphocytes	74.76 ± 3.2	89.21 ± 2.54 ***	62.92 ± 1.22 ***
5	Neutrophils	15.17 ± 1.72	24.95 ± 1.93 ***	17.22 ± 1.04 ***
	Eosinophils	9.27 ± 1.17	9.04 ± 1.20	9.10 ± 0.33
	Monocytes	2.17 ± 0.29	2.28 ± 1.39	2.21 ± 0.20

Data are expressed at mean ± SD (n=6), significantly different at *p<0.05, **p<0.01 and ***p<0.001 in comparison to control group. Control group - Polysorbate-80 (2%), Deltamethrin - 0.5% and AH - Arecoline hydrobromide (12.5 mg/mL); Hb = hemoglobin, PCV = packed cell volume, TEC = total erythrocyte count, TLC = total leukocyte count, DLC = differential blood count.

was 9.01 g/100 mL, PVC was 29.24 %, TEC and TLC were 5.23 and 7.19 10⁶/cumm, respectively in comparison to the control group at 144 h. However, the hematological values were slightly disturbed in the reference group after treatment with the deltamethrin. Results exhibited that rise in TEC was achieved in arecoline treated group post-treatment. Neutrophil count was significantly raised (17.22 %) in arecoline treated group. Monocyte count was also slightly increased (2.21 %) after treatment with arecoline hydrobromide. However, other values were indifferent as compared to the control group.

3.4. Skin irritation assay

The application of arecoline hydrobromide emulsifiable concentrate in the concentration of 12.5 mg/mL on calves skin showed mild erythema after 20 min of treatment. However, it was disappeared after 6 h onwards to 48 h of observation. The erythema score was 1 at 20 min but the score was 0 at 6 h, 12 h, 24 h, and 48 h. No redness or reaction was seen to the control animals with polysorbate-80 (2%). Interestingly, no edema was seen in any animals.

4. Discussion

Cattle have a paramount role in agriculture in the developing countries where tick and tick-borne diseases affecting tremendously to earn their livelihood (De Meneghi et al., 2016). In the present study, the topical application of arecoline hydrobromide emulsifiable concentrate was evaluated on calves infested with *R. microplus*. In our previous work on evaluation of *in vitro* efficacy of arecoline hydrobromide, significant acaricidal activity of the drug was seen on both *R. microplus* adults and larvae with the conclusion that the acaricidal activity of drug may be due to its oxidative stress inducing activity. The study also showed that arecoline hydrobromide was effective in reducing oviposition, hatchability, reproducibility and egg weight of female ticks (Jain et al., 2020d). Isolates of herbal extracts and essential oils possess significant acaricidal activity against cattle ticks (Baran et al., 2020; Krishna et al., 2014). Herbal acaricides can replace the synthetic one for which ticks have been developing resistance. It can also reduce the negative impact of synthetic chemicals residues in tissues, bad effects on the ecosystem and on humans health. The toxicity study must be performed to check the safety concern of drugs before evaluation biological activity (Jain et al., 2016, 2015; Jain et al., 2014; Sharwan et al., 2015).

in vivo acaricidal activity of arecoline hydrobromide emulsifiable concentrate was evaluated on red Sahiwal calves. Results showed that application of arecoline on calves skin caused a reduction of ticks number as compared with the control group. Our results were in agreement with the findings of Chen et al., 2019. The isolated herbal compounds such as thymol, carvacrol and p-cymene from oregano oil showed acaricidal activity in the lower concentration i.e. 1% and 5% (Shang et al., 2016). AH showed no toxicity in acute dermal toxicity study on wistar rats. In the present study, the topical application of arecoline hydrobromide showed only slight and transient erythema on animal's skin after 20 min. The liquid formulations in the form of emulsion, suspension or solution applied topically on animal's skin may also promote absorption of the drug into the bloodstream thus providing protection from both internal and external parasites (Pitman and Rostas, 1981). The injectable formulations of chemicals are labor-saving but posing a detrimental effect on animals' health (Schröder, 1992). In this respect, plant-derived bioactive compounds may be very effective as ectoparasiticides due to their properties like biodegradability and eco-friendly nature (Wanzala, 2017; Adenubi et al., 2016; Khare et al., 2019). However, while the topical dosage form is not intended to deliver drugs into the systemic circulation, it may do so to the animals under abnormal conditions while animals dipping in the tanks. The topical application of herbal drug on cattle shows significant acaricidal activity (Jain et al., 2020e; Zaman et al., 2012). Instead of the old method of application of acaricidal formulations as topically but it is safe

to the animals and does not cause undesired systemic side effects to the animals.

Furthermore, the hematological profile of calves including number of eosinophils, lymphocytes, neutrophils and other cells count were not significantly different in treatment group as compared to control group. Reduced hemoglobin and total erythrocyte count are indications of leucocytosis in animals due to anemia produced by blood-sucking ticks and hemorrhage (Raut et al., 2008). The hematological profiles such as Hb, PCV, and RBCs decrease significantly in infested cattle as compared to control, however, WBCs increases significantly in infested animals (Papadopoulos, 2013). In the present study, spraying of arecoline hydrobromide on calves skin significantly reduced tick burden along with improving red blood cell parameters.

Arecoline is a cholinergic agonist that increases the concentration of acetylcholine in nervous tissue and is responsible for pupillary constriction (Murphy et al., 1984). The acaricidal activity of arecoline may be due to its oxidative stress inducing property on ticks. Inhibition of acetylcholinesterase may leads to rise the level of acetylcholine in ticks that results into paralysis and finally death (Jain et al., 2020c). Results showed that arecoline hydrobromide was effective in *in vivo* assays with minimal toxicity to animals while applying topically.

5. Conclusion

Arecoline hydrobromide showed significant acaricidal activity against the cattle tick *R. microplus* along with clinical safety to the animals. It was found effective on topical application to the animals with a mild irritancy effect on skin. The skin lesions were restored after the application of arecoline topically with a reduced number of ticks on skin. It can be concluded from the study that arecoline may be used as an alternative herbal acaricidal drug to existing synthetic acaricidal drugs for which the ticks have already produced resistance. Thus, topical application of arecoline hydrobromide may keep animals healthy and tick free without affecting the ecosystem and human beings.

CRedit authorship contribution statement

Parag Jain: Conceptualization, Methodology, Writing - original draft, Writing - review & editing. **Trilochan Satapathy:** Data curation, Supervision. **Ravindra Kumar Pandey:** Supervision.

Declaration of Competing Interest

The authors report no declarations of interest.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.vetpar.2021.109490>.

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